MILITARY SPECIFICATION

RELAY, ARMATURE, TYPE MA-1

1. SCOPE

1.1 This specification covers one type of normally closed, single pole, 115 volt, 400 cycle alternating current relay, designated Type MA-1.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND PUBLICATIONS

2.1 The following specifications, standards, and publication, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-14	Plastic-Materials, Modeling, and Plastic-Parts, Molded; Thermosetting
MIL-D-5028	Drawings and Data Lists; Preparation of (For Engines, Accessories and Other Auxiliary Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, (General Specification For)
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-S-7742	Screw Threads; Standard, Aeronautical
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts

STANDARDS

MILITARY

MIL-STD-129	Marking of Shipments

AIR FORCE-NAVY AERONAUTICAL

AN960	Washer - Flat

PUBLICATION

AIR FORCE-NAVY AERONAUTICAL BULLETIN

143	Specifications and Standards; Use of

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Component parts. - The relay assembly shall consist of an undervoltage sensing relay, housing, and necessary connections.

3.2 Specifications and standards. - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin No. 143, except as provided in the following paragraph:

3.2.1 Standard parts. - AN or MIL Standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL Standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.2.2 Protective treatment. - When materials are used in the construction of the relay that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

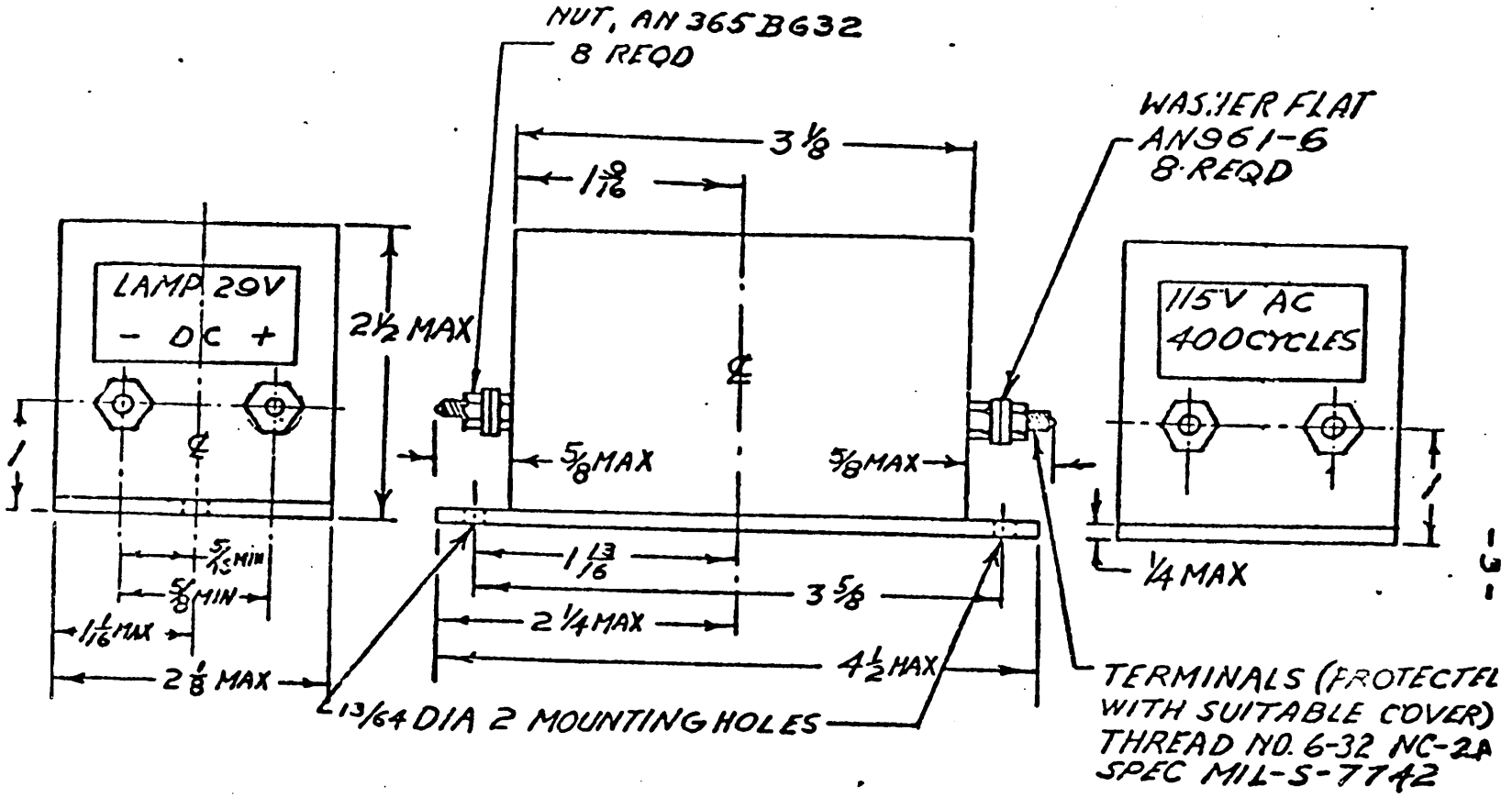
3.3 Design. - The relay shall be designed to make a contact which will energize a failure warning light whenever a voltage of less than 92.5 ±2.5V ac is sensed by the relay. The contact shall remain closed until the voltage sensed by the relay increases to 102.5 ±2.5V ac, at which voltage it shall open and remain open until the voltage is again reduced to 92.5 ±2.5V ac.

3.3.1 Frequency. - The relay shall be designed to withstand frequencies of 320 to 1200 cps without damage. Calibration of the relay for test purposes, however, shall be at 400 ±10 cps.

3.4 Construction. - The relay shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 The relay shall be constructed substantially in accordance with Figure 1.

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERENCES: FRACTIONS ± 1/64

DIMENSIONS OF TERMINALS CONSISTANT FOR BOTH ENDS

FIGURE 1. OUTLINE AND MOUNTING DIMENSIONS

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3.4.2 The relay shall be constructed to insure proper operation when mounted in any position, and the complete mechanism shall be totally enclosed for mechanical, sand, and dust protection.

3.4.3 Interior mechanism. - The interior mechanism of the relay shall be constructed to operate satisfactorily for 1000 hours of service usage without adjustment.

3.4.4 Holding coil:

3.4.4.1 Winding. - The relay holding coil shall be properly insulated and taped. It shall be able to withstand an operating voltage of 130V ac applied to its terminals for 3 minutes without damage to the insulation.

3.4.4.2 Heads. - The relay holding coil heads, if used, shall be of phenolic material conforming to Specification MIL-P-14, or equivalent.

3.4.4.3 The alternating current relay (voltage sensing) shall be designed to pick up at 102.5 ±2.5V ac and to drop out at 92.5 ±2.5V ac under all conditions relative to aircraft use as specified herein.

3.4.5 Contact base. - The contact base shall be of phenolic material conforming to Specification MIL-P-14.

3.4.6 Contact alignment. - The movable contacts of the relay shall be self aligning with the stationary contacts and shall be positive in operation.

3.4.7 Spacing and creepage distances. - The minimum spacing between current carrying parts and other portions of the relay, other than suitable insulating material, and also between current carrying parts of opposite polarity shall be not less than 1/16 inch.

3.4.8 Rating. - The rating of the relay shall not be less than 3 amp at 29V dc.

3.4.9 Terminal posts:

3.4.9.1 Both leads of the relay holding coil shall be brought out on one end of the relay to terminal posts and insulated from the coil housing. The leads shall not be exposed to possible damage in service. The load terminal posts shall be attached to the opposite end of the relay and insulated from the coil housing.

3.4.9.2 The coil and load terminal posts shall be threaded with No. 6-32 NC-2A threads, and shall be provided with the proper nuts and lock washers or locknuts for attaching the cable terminals. The terminals shall not be exposed to possible damage in service and shall have a minimum thread allowance of 1/8 inch.

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3.4.9.3 The terminals shall be securely fastened to the control so that the terminals cannot be twisted, pushed in, or pulled out in normal service use.

3.4.10 Wiring:

3.4.10.1 The internal wiring of the relay shall be with wire capable of withstanding the insulation tests and shall be of a suitable gage to withstand the tests specified in Section 4.

3.4.10.2 The wires shall be arranged and secured to prevent the cable from interfering with the proper operation of any of the components of the relay.

3.4.11 Container. - The entire relay assembly may, at the discretion of the manufacturer, be enclosed in a leakproof metal container and pressurized with an inert gas or blend of gases with a minimum of 10 percent helium by volume to a pressure greater than sea level atmospheric pressure. Leakage of the pressurizing gas shall not exceed one micron cubic foot per hour. The exterior of the sealed container shall be suitably protected from corrosion in a salt fog atmosphere by painting or plating as required.

3.5 Performance. - The relay shall be capable of satisfactory operation under the following conditions:

3.5.1 Temperature. - At ambient temperatures ranging from -65 to +71°C.

3.5.2 Altitude. - A pressure altitude range from sea level to 65,000 feet.

3.5.3 Sand and dust. - Exposure to airborne sand and dust particles encountered in normal desert operation.

3.5.4 Salt spray. - Exposure to atmosphere containing salt-laden moisture.

3.5.5 Vibration. - Vibration incident to service use.

3.5.6 Humidity. - All conditions of relative humidity.

3.6 Interchangeability. - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.7 Screw threads. - Screw threads shall be in accordance with Specification MIL-S-7742.

3.8 Weight. - The weight of the relay shall not be more than 1.2 pounds.

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ible to main contacts only). Relays shall make and break 4.7.20 Rupture (see 3.5.25) (double

3.9 Dimensions. - The dimensions of the relay shall be substantially as shown on Figure 1.

3.10 Finish. - All exposed steel parts shall be of corrosion-resistant steel or protected by a plating of corrosion-resistant metal.

3.11 Identification of product. - The relay shall be marked for identification in accordance with Specification MIL-M-7911. The nameplate shall contain the following information:

Relay, Armature
(115V AC, 400 Cycles Single Pole)
Type MA-1
Specification MIL-R-7611
Mfr's. Part No._____
Mfr's. Serial No._____
Order No._____
Stock No._____
(Mfr's. Name or Trademark)
U. S. Property

3.12 Workmanship:

3.12.1 General. - The relay, including all parts and accessories, shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, accuracy of dimensions, and freedom from burrs and sharp edges.

3.12.2 Cleaning. - The relay shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests. - The inspection and testing of relays shall be classified as follows:

4.1.1 Qualification tests. - Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

4.1.2 Inspection tests. - Inspection tests are those tests accomplished on relays submitted for acceptance under contract.

4.2 Test conditions:

4.2.1 Standard atmospheric conditions. - Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

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4.3 Qualification tests:

4.3.1 Prior qualification. - Unless otherwise specified by the Procuring Agency, relays which have not previously passed a qualification test, or which have passed the qualification test and have been modified in any manner shall satisfactorily pass a qualification test prior to the acceptance of any relay.

4.3.2 Qualification test samples. - Qualification test samples shall consist of 3 relays. Samples shall be appropriately identified with the manufacturer's own part number and any additional identification required by the authorizing letter:

4.3.3 The qualification tests shall consist of the following tests and those specified under Inspection tests and shall be conducted in the following order:

Relay No. 1

1. Coil characteristics
2. Humidity
3. Sand and dust
4. Life test

Relay No. 2

1. Coil characteristics
2. High temperature
3. Low temperature
4. Altitude
5. Vibration resistance
6. Acceleration resistance
7. Salt spray
8. Life test

Relay No. 3

1. Coil characteristics
2. Fungus
3. Life test

4.3.3.1 Coil characteristics:

4.3.3.1.1 Opening voltage. - The relay shall be subjected to test to determine that the main contacts open with an alternating current potential of 102.5 ±2.5V ac applied to the holding coil terminals and to determine that the main contacts remain open on 92.5 ±2.5V when the opening voltage applied to the holding coil is reduced from 115V.

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4.3.3.1.2 Operating voltage. - The relay shall be subjected to an alternating current potential of 130V applied to the holding coil terminals for a 3-minute period. At the completion of this test, the relay shall be subjected to the Insulation test specified in 4.4.2.2.1.

4.3.3.1.3 Continuous operation. - The relay shall be subjected to 5 hours continuous operation with 115V applied to the holding coil. At the completion of this period, the main contacts shall remain open on 92.5 ±2.5V when the applied voltage is reduced from 115V.

4.3.3.2 Current consumption. - The relay shall be tested to determine that the holding coil consumes not more than 0.06 amp when operating on 115V ac.

4.3.3.3. Vibration resistance. - The relay shall be subjected to vibration in accordance with Procedure I of Specification MIL-E-5272. A test lamp or other suitable indicating device shall be connected across the main contacts while the relay is undergoing this test to determine that the vibration does not open the contacts with 92.5 ±2.5V across the coil or close them with 102.5 ±2.5V across the coil. The relay contacts shall satisfactorily open during this test with 102.5 ±2.5V across the coil and stay open as low as 92.5 ±2.5V.

4.3.3.4 Acceleration resistance. - The relay shall be subjected to acceleration forces of 10 g. The time required to attain the 10 g acceleration shall not be more than 2 seconds. The relay shall be mounted in the position most likely to cause malfunctioning, and shall be accelerated for a minimum of one minute without causing the main contacts to open with no voltage across the coil or close when the coil voltage is reduced to 92.5 ±2.5V on the relay. A test lamp or suitable indicating device shall be connected across the main contact terminals, while the relay is undergoing this test, to determine that the acceleration does not result in closing or opening of the contacts.

4.3.3.5 Low temperature. - The relay shall be subjected to -65°C for a minimum period of 16 hours. At the end of this period and while at the specified temperature, the relay shall be subjected to the test specified in 4.3.3.2. The current consumed shall not exceed that specified. The relay shall then be tested intermittently for closing and opening voltages until it attains a temperature of 20 to 30°C. The voltage characteristics during this test shall conform to the applicable test specified in 4.3.3.1.1.

4.3.3.6 High temperature. - The relay shall be subjected to a temperature of +71°C for a minimum period of 16 hours. At the completion of this period and with the relay at the specified temperature, the relay shall be subjected to the applicable test specified in 4.3.3.1.1. The relay shall then be tested intermittently for closing and opening voltages until it attains a temperature of 20 to 30°C. The voltage characteristics during this test shall conform to the applicable test specified in 4.3.3.1.1

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4.3.3.7 Life test. - The relay shall be subjected to test for making and breaking a 3 amp resistive load at 29.0V dc for 40,000 operations. The direct current load shall be "made" by connecting 90V ac 400 cycles to the alternating current terminals and the direct current load shall be broken when 105V ac is applied to the alternating current terminals. The operations shall be conducted at the rate of 60 cpm with an approximate 0.5 second "on" and "off" period. The total of 40,000 cycles may be accumulated by intermittent operation, however, no adjustments shall be made on the relay during any part of the Qualification test.

4.3.3.8 Environmental. - The following tests shall be conducted in accordance with the specified procedures of Specification MIL-E-5272 and as specified herein:

4.3.3.8.1 Humidity. - Procedure I. Immediately after this test, the relay shall satisfactorily pass the applicable test specified in 4.3.3.1.1 and 500 continuous cycles in accordance with the test specified in 4.3.3.7.

4.3.3.8.2 Sand and dust. - Procedure II for a period of 5 hours.

4.3.3.8.2.1 After subjection to the sand and dust test, the relay shall be subjected to the following tests in the sequence specified, immediately after attaining a room temperature of 20 to 30°C:

(1) Opening voltage test as specified in 4.3.3.1.1.
(2) Five hundred continuous cycles as specified in 4.3.3.7.

4.3.3.8.3 Fungus. - Procedure I. No fungus growth which will affect subsequent operation shall result from this test.

4.3.3.8.4 Salt spray. - Procedure I. At the completion of a 50-hour period, the relay shall satisfactorily pass the tests specified in 4.3.3.1.1 and 4.3.3.7.

4.3.3.8.5 Altitude. - Procedure II. The relay shall be subjected to an ambient temperature of -65°C for a period of 4 hours. At the end of this period and while still at the required temperature and altitude, the relay shall satisfactorily pass the test specified in 4.3.3.1.1.

4.4 Inspection tests:

4.4.1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record of report of inspection and tests shall be signed or approved by a responsible person specifically assigned

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by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting te ts to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.4.2 Individual tests. - Each relay shall be subjected to the following tests:

4.4.2.1 Examination of product. - Each relay shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.2.2 Insulation test. - The relay shall be subjected to the following voltages between the appropriate terminals and other exposed metal parts:

4.4.2.2.1 One thousand two hundred and fifty volts rms at commercial frequency between each terminal and other exposed metal parts for 60 seconds. There shall be no failure as a result of this test.

4.4.3 Relays shall be selected at random from each lot in accordance with the following table and subjected to the following tests:

Lot size	Sample size
1 - 500	1
501 - 1500	2
over 1500	3

4.4.3.1 Lot. - A lot shall consist of relays manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.3.2 When sampling tests are specified on a number of items that are selected as representative of a certain lot, and one or more of this number fails to meet the specified test(s), acceptance of all items shall be withheld until the extent and cause of failure is determined. For operational reasons, individual tests may be continued pending investigation of a sampling test failure, but the final acceptance of the product is contingent upon the Inspector's decision regarding the overall conformance of the product to specification requirements. If investigation indicates that the defects may exist on items previously accepted, full particulars concerning the defect(s) found, including recommendations for correction, will be furnished to the Procuring Agency.

4.4.3.3 Weight. - The relay shall be weighed and shall not exceed the weight specified in Section 3.

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4.4.3.4 The relay shall be subjected to the following tests specified in the Qualification test:

(1) Opening voltage
(2) Resistive load test (life test, except for 500 cycles only)
(3) Vibration resistance
(4) Life test

4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with the specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 General. - The packaging, packing, and marking requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

5.2 The relay shall be packaged and packed for shipment in accordance with Group V of Specification MIL-P-6064, in a Type II, Class I container.

5.3 Marking of shipments. - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Relay, Armature, 115V 400 Cycle AC, SPST 3 Amp 29V Contacts, Type MA-1, (*Mfr's. Part No._____), Specification MIL-R-7611.

* Applicable data to be entered by the contractor.

In addition the following data shall also be included:

IF IN STORAGE AFTER (DATE) (1), THE RELAY SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

(1) This date shall be 18 months from the date of inspection.

6. NOTES

6.1 Use. - The Type MA-1 relay covered by this specification is intended for use in aircraft as a means of remotely controlling the operation of an inverter failure warning light for indicating inverter undervoltage conditions (below 90V ac).

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6.2 Qualification tests. - In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on a Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Wright Air Development Center and arrange to have the products that they propose to offer to the Air Force tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commanding General, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, Attention: WCLE.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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